

WebFundamentals

Web Development Overview

1. What is Web 3.0 and how is it different from Web 1.0 and Web 2.0?

Web 1.0 was the first stage of the web. Most websites were simple and mainly used for reading information. Users could visit a page and look at the information, but there was not much they could do on the website.

Web 2.0 made websites more interactive. People could not only read information but also create and share their own content. Social media, online comments and websites where people work together are good examples of Web 2.0.

Web 3.0 is the next idea of how the web can develop. It is often described as a web that is more intelligent, connected and decentralised. It can use technologies such as artificial intelligence and blockchain, and it aims to give users more control over their information and digital assets.

So, the main difference is that Web 1.0 was mostly about reading, Web 2.0 is about reading and interacting, and Web 3.0 is focused on a smarter and more connected web where users can have more control.

2. What are the functional differences between the front end and back end of a web application?

The front end is the part of a website that I can see and use. It includes things like buttons, menus, forms, images, text and the overall layout of the website. HTML, CSS and JavaScript are some of the technologies used to build the front end.

The back end works in the background. It is responsible for things such as processing information, handling requests, applying the application's logic and communicating with databases. The user normally does not see the back-end code.

For example, if I log into a website, the login form is part of the front end. When I enter my details and click the login button, the back end receives the information and checks whether the details are correct. It then sends a response back to the front end.

3. From typing a URL to seeing a web page

When I type a URL into my browser, the browser uses the address to find the web resource I am asking for. The browser acts as the client and sends an HTTP request to the web server.

The web server receives the request and works out what needs to be returned. Sometimes the server needs to use back-end code or get information from a database before it can respond.

The server then sends an HTTP response back to my browser. The response normally contains a status code, headers and a body. The browser receives this response and uses the information to display the web page.

A simple example of an HTTP request is:

```
GET /index.html HTTP/1.1
Host: example.com
Accept: text/html
```

If the request is successful, the server could send a response such as:

```
HTTP/1.1 200 OK
Content-Type: text/html

<html>
  <body>Hello World</body>
</html>
```

The 200 OK means that the request was successful. The Content-Type header tells the browser what type of content it has received. The body contains the actual content that the browser can use to display the page.